Graphing Proportional Relationships
Answer Key

Grades 6–7 Ratios and Proportional Relationships

Quick Answers

1. 8, 8	4. 2.5	7. 3, 2	10. 18
2. 2.5, 2.5	5. 8	8. 16, 4	
3. 3	6. 5, 4, 3.5	9. 3.5, 42	

Common wrong answers

3. *If they got 0.333333: read the point backwards and divided time by distance*

1. Dog-walking pay: plot the table and find the unit rate.

Step 1. The three points are $(0, 0)$, $(3, 24)$ and $(5, 40)$. Plotted, they lie on one straight line, and that line starts at the origin — the mark of a proportional relationship.

Step 2. Unit rate from the origin to $(3, 24)$: $k = \dfrac{24-0}{3-0} = 8$.

Step 3. Check between two points that are not the origin: $k = \dfrac{40-24}{5-3} = \dfrac{16}{2} = 8$. The same rate everywhere is what makes the graph a straight line through $(0, 0)$.

Manual review: the plotted points and the drawn line are the student's own work — check that all three points are placed correctly and the line is ruled through the origin. Dev earns 8 dollars per hour. $\boxed{k = 8}$

2. Rice cost: plot the table and find the cost of one kilogram.

Step 1. Points $(0, 0)$, $(2, 5)$ and $(6, 15)$. All three lie on one line through the origin, so cost is proportional to mass.

Step 2. From the origin to $(2, 5)$: $k = \dfrac{5}{2} = 2.5$.

Step 3. Between the two non-origin points: $\dfrac{15-5}{6-2} = \dfrac{10}{4} = 2.5$ — the same constant.

Manual review: inspect the student's plotted points and ruled line. One kilogram of rice costs 2.5 dollars. $\boxed{k = 2.5}$

3. Cyclist graph: unit rate in kilometres per hour.

Step 1. The line goes through the origin, so the unit rate is $y \div x$ at any point on it. Use the marked point $(5, 15)$.

Step 2. $k = \dfrac{15}{5} = 3$ kilometres per hour.

Common wrong answer: 0.33, from dividing hours by kilometres. Always put the quantity named by the y-axis on top. $\boxed{k = 3}$

4. Paint graph: write the equation.

Step 1. A line through the origin has the form $y = kx$, so the only thing to find is k.

Step 2. The marked point is $(4, 10)$, so $k = \dfrac{10}{4} = 2.5$ square metres per litre.

Step 3. Substituting the constant gives the equation $y = 2.5x$. $\boxed{y = 2.5x}$

5. Barrel: how long to collect 12 litres.

Step 1. The equation is $y = 1.5x$, and the question fixes the output at $y = 12$, so solve $1.5x = 12$ for x.

Step 2. $x = \dfrac{12}{1.5} = 8$ minutes.

Step 3. The graph agrees: reading across from 12 litres meets the line above 8 minutes.

$\boxed{x = 8}$

6. Is the relationship proportional?

Step 1. Divide y by x in every column. If the relationship is proportional, all the quotients are the same number.

Step 2. $5 \div 1 = 5, \quad 8 \div 2 = 4, \quad 14 \div 4 = 3.5.$

Step 3. The three quotients are different, so there is no single constant of proportionality. The graph settles it too: the line is straight but it meets the y-axis at 2, not at the origin, so it is not a proportional relationship. $\boxed{5,\ 4,\ 3.5:\ \text{not proportional}}$

7. Two runners: both unit rates and who is faster.

Step 1. Both lines pass through the origin, so each unit rate is $y \div x$ at that line's marked point.

Step 2. Runner A through $(4, 12)$: $\dfrac{12}{4} = 3$ metres per second. Runner B through $(5, 10)$: $\dfrac{10}{5} = 2$ metres per second.

Step 3. The steeper line has the larger unit rate, so runner A is faster — by 1 metre per second. $\boxed{A = 3,\ B = 2}$

8. Water tank: difference between two readings, and the unit rate.

Step 1. Read the marked points: at 1 minute the tank holds 4 litres, at 5 minutes it holds 20 litres.

Step 2. Difference: $20 - 4 = 16$ litres more.

Step 3. Unit rate: $\dfrac{20-4}{5-1} = \dfrac{16}{4} = 4$ litres per minute, which matches $y \div x$ at either point. $\boxed{16,\ k = 4}$

9. Bottling machine: unit rate, then a prediction off the graph.

Step 1. The marked point is $(4, 14)$ and the line goes through the origin, so $k = \dfrac{14}{4} = 3.5$ hundreds of bottles per hour.

Step 2. The equation is $y = 3.5x$.

Step 3. At $x = 12$ hours: $y = 3.5 \times 12 = 42$ hundreds of bottles. The drawn axis stops at 6 hours, which is exactly why the equation is worth having. $\boxed{k = 3.5,\ y = 42}$

10. Printer: draw the graph, then find the time for 45 pages.

Step 1. 5 pages every 2 minutes is a unit rate of $\frac{5}{2} = 2.5$ pages per minute, so the equation is $y = 2.5x$ and the graph is the line through $(0, 0)$ and $(20, 50)$.
Step 2. Set the output to 45 pages: $2.5x = 45$.
Step 3. $x = \frac{45}{2.5} = 18$ minutes. On the graph, 45 pages sits between the gridlines at 40 and 50, above 18 minutes.
Manual review: the drawn line is the student's own work — it must start at the origin and pass through $(4, 10)$ and $(20, 50)$. $\boxed{x = 18}$